

MATTHEW VAISHNAV

PROFESSIONAL SUMMARY

Production machine learning engineer with expertise in medical AI systems and healthcare infrastructure. Architected HistoCore, a comprehensive computational pathology platform achieving **93.98% AUC** with **8-12x training optimization** and **1,483-test validation suite**. Demonstrated ability to bridge research innovation and production deployment, delivering systems that process **262K+ clinical samples** with **<5 second inference** and **90% cancer detection sensitivity** - directly improving patient outcomes through measurable clinical impact.

EDUCATION

TECHNICAL PROJECTS

HistoCore — Production Medical AI Platform | *2026*

- Achieved **93.98% test AUC** and **84.26% accuracy** on PatchCamelyon dataset (262K training, 32K test samples)
- Delivered **90% cancer detection sensitivity** reducing missed diagnoses by **61.7%** in clinical screening scenarios
- Pioneered **6-13x training optimization** breakthrough reducing computational requirements from 20-40 hours to **3.1 hours**
- Architected **24/7 production infrastructure** with Prometheus monitoring, Grafana dashboards, and automated incident response
- Built **comprehensive validation framework** with **1,483 tests** (55% coverage) including property-based testing with Hypothesis
- Achieved **<5 seconds inference time** with ONNX optimization and horizontal scaling via Docker/Kubernetes
- Engineered **PACS integration** supporting multi-vendor systems (GE/Philips/Siemens/Agfa) with DICOM C-FIND/C-MOVE/C-STORE
- Implemented **HIPAA-compliant architecture** with 7-year audit retention, tamper-evident logging, and AES-256-GCM encryption
- Deployed **hospital workflow integration** with Epic, Cerner, and Allscripts EMR systems for seamless clinical adoption
- Developed **federated learning framework** with $\epsilon \leq 1.0$ differential privacy, DP-SGD implementation, and Byzantine-robust aggregation
- Built **privacy-preserving multi-site training** enabling collaboration across hospital networks without data sharing

- Implemented **real-time monitoring system** with convergence detection, privacy budget tracking, and automated alerting

SENTINEL — Enterprise DDoS Protection System | *2025-2026*

- Achieved **96% accuracy** and **98.33% recall** on CIC-DDoS2019 industry benchmark dataset
- Engineered **async worker architecture** maintaining stable response times under 100K+ requests/second volumetric attacks
- Implemented **dynamic threat detection** with 96.41% precision minimizing false positives on legitimate traffic
- Built **WebSocket P2P mesh** enabling millisecond threat propagation across regional instances without central authority
- Designed **horizontal Redis clustering** for cross-region state synchronization and instant IP reputation lookups
- Created **proof-of-threat consensus** mechanism eliminating single points of failure in distributed deployments

Drift — Infrastructure State Management | *2026*

- Built **cross-platform CLI tool** (Linux/macOS/Windows) for configuration change tracking
- Implemented **rollback capabilities** and known-good state management for production environments
- Designed **tamper-evident logging** with cryptographic integrity verification

PROFESSIONAL EXPERIENCE

- Executed fundraising event planning and donor market research initiatives
- Contributed to multi-platform social media campaigns driving community engagement
- Managed operational workflows for customer experience optimization
- Performed facility maintenance and hardware repairs ensuring 99%+ uptime
- Created social media content driving customer acquisition across X/Twitter and Facebook

TECHNICAL EXPERTISE

PyTorch, Computational Pathology, Multiple Instance Learning (MIL), Foundation Models, Federated Learning, Model Optimization, ONNX, Property-Based Testing (Hypothesis)

Docker, Kubernetes, Terraform, Ansible, GitHub Actions CI/CD, Prometheus, Grafana, Redis Clustering, Load Balancing, Incident Response

DICOM/FHIR/PACS Integration, HIPAA Compliance, Medical Image Processing, Whole-Slide Imaging (WSI), OpenSlide, Clinical Validation, FDA Regulatory Pathways

Security Onion, Suricata, Zeek, pfSense, Wireshark, SAST/DAST, Container Security (Trivy), Network Security, VLAN Design, Active Directory

Python, JavaScript (Node.js), C++, Bash, PowerShell, SQL, Redis, PostgreSQL, RESTful APIs, WebSocket, Async Programming

AWS (VPC, GuardDuty, CloudWatch, S3), Azure Sentinel, Linux (Ubuntu, Debian), Windows Server, VMware, Mobile Development (React Native)

CERTIFICATIONS & ACHIEVEMENTS

- CompTIA Security+ (62% complete)
- TryHackMe SOC Level 2 (78% complete)
- TryHackMe Pre-Security
- Cisco Networking Essentials
- First Aid Certification (March 2024 – Present)
- WHMIS 2015 Certified
- Royal Conservatory of Music Piano Level 3 (Honors and Distinction)

KEY ACCOMPLISHMENTS

- **Reduced missed cancer diagnoses by 61.7%** through optimized sensitivity thresholds (90% vs baseline 77.9%)
- **Pioneered 6-13x training optimization** breakthrough enabling consumer GPU deployment of medical AI systems
- **Developed comprehensive federated learning** framework with formal differential privacy guarantees ($\epsilon \leq 1.0$)
- **Architected 24/7 production platform** processing 262K+ clinical samples with <5 second inference latency
- **Built industry-leading validation suite** with 1,483 tests including property-based testing for correctness guarantees
- **Achieved 96% threat detection accuracy** on industry benchmark dataset with real-time response capabilities
- **Engineered PACS integration** supporting all major hospital vendors (GE/Philips/Siemens/Agfa) with HIPAA compliance
- **Implemented privacy-preserving ML** enabling multi-site hospital collaboration without patient data sharing
- **Delivered production-ready systems** with comprehensive monitoring, incident response, and automated recovery